

Client-Server

Client–Server Systems

Learning Objectives

- Be able to describe the client-server model and protocol stack
- Be able to describe and use serialization to send data between machines

Client-Server Overview

- Real systems are distributed:
 - Browsers and web servers
 - mobile apps and backend servers
 - Services to services.
- Consequently, processes on different hosts must “speak” to each other!
- Client-server design:
 - Client: a process that sends requests to ask for service
 - Server: a process that receives requests to provide services.
 - Protocol: the mechanism that machines use to communicate
 - See recording for overview of typical timeline

The TCP/IP protocol stack

- The key technology stack underlying the modern internet:
 - Application: An application’s format and protocol (HTTP, JSON, protobuf, ...)
 - Agents involved are processes
 - Transport: Ensures end-to-end host communication (TCP, UDP)
 - Agents involved are processes
 - Network: IP addressing and routing (IPv4, IPv6)
 - Agents involved are routers, hosts, etc.